

LaunchLens AI — Data Quality Report

Startup/company dataset built for the ML/NLP component of the LaunchLens AI major project
Report generated: 2026-08-13

1. Summary

This report documents a real, source-verifiable startup/company dataset built entirely from three publicly-posted sources (see Sources sheet / section 6). The full cleaned pool of real, deduplicated records is 67,021. The main deliverable, LaunchLens_Master_Dataset, is a reproducible stratified sample of 5,013 records drawn from that pool (stratified by industry and outcome so rare classes like IPO and Closed remain represented; random_state=42). No company, funding figure, investor, or outcome in this dataset was invented — fields the source data does not contain are explicitly marked Unknown rather than filled in.

2. Required Report Metrics

Metric	Value
1. Total records (main deliverable)	5,013
Total records (full cleaned pool, bonus file)	67,021
2. Unique companies (main deliverable)	5,011
3. Active / Ongoing	4,006
4. Acquired	413
5. IPO	121
6. Failed / Closed	463
7. Unknown outcome	10
8. Countries represented	76
9. Industries represented	22
10. Average total_funding_usd (where known)	\$15,782,348
Median total_funding_usd (where known)	\$2,000,000
12. Exact duplicates removed in cleaning	102
Possible duplicates flagged (not removed)	23
14. Founded-year coverage (main deliverable)	1910–2015

11. Source Distribution

Source	Records
Kaggle: big-startup-secsees-fail-dataset-from-crunchbase	4,924
Kaggle: startup-investments-crunchbase	64
User-supplied: Startups & Funding Dataset 2024-2025	25

13. Missing Value % by Column (main deliverable)

Column	Missing %
company_description	100.0%
target_audience	100.0%
customer_type	100.0%
problem_solved	100.0%
solution_description	100.0%
technology_used	100.0%
business_model	100.0%
revenue_model	100.0%
pricing_model	100.0%
revenue_stream	100.0%
employee_count	100.0%
latest_known_valuation_usd	100.0%
acquisition_date	100.0%
ipo_date	100.0%
shutdown_date	100.0%
last_funding_amount_usd	99.5%
major_investors	99.5%
success_label	80.1%
business_category	31.1%
founded_year	22.3%
total_funding_usd	18.9%
headquarters_city	12.3%
founded_country	10.3%
headquarters_country	10.3%
geographic_market	10.3%
competitor_count	10.3%
website	8.1%
sub_industry	4.7%
funding_rounds	0.5%
startup_id	0.0%
startup_name	0.0%
industry	0.0%
market_size_category	0.0%
market_growth_category	0.0%
demand_level	0.0%
competition_level	0.0%

last_funding_date	0.0%
startup_stage	0.0%
startup_status	0.0%
acquisition_status	0.0%
ipo_status	0.0%
success_indicator	0.0%
outcome_reason	0.0%
data_quality	0.0%
source	0.0%
source_url	0.0%
last_verified_date	0.0%
possible_duplicate_flag	0.0%

14. Success Label Methodology

SUCCESS LABEL METHODOLOGY (documented per project requirement #4)

Source field: `status` in the Crunchbase-derived Kaggle exports, one of:
operating | acquired | closed | ipo

Mapping to success_indicator (categorical):

status == 'ipo' -> "IPO"

status == 'acquired' -> "Acquired"

status == 'closed' -> "Failed / Closed"

status == 'operating' -> "Active / Ongoing" (NOT labeled "successful")

status missing/other -> "Unknown"

Mapping to success_label (binary ML target):

1 = IPO OR Acquired (status == 'ipo' or 'acquired')

0 = Failed / Closed (status == 'closed')

NaN (NULL) = Active/Ongoing OR Unknown status

Rationale: an "operating" flag only tells us the company had not yet closed, IPO'd, or been acquired as of the Crunchbase snapshot date. It is NOT evidence of success (many long-running "operating" companies are marginal, and many successful acquisitions/IPOs happened to companies that were merely "operating" the day before). Per the project's explicit instruction, we do NOT classify an active company as successful merely for being active. Records with insufficient evidence keep success_label = NaN so they can be correctly excluded from binary classification training rather than silently miscounted as failures (0) or successes (1).

KNOWN LIMITATION: the source Crunchbase snapshot exports are not continuously updated - a company shown as "operating" may have IPO'd, been acquired, or shut down AFTER the snapshot date and simply not been recaptured. This is a limitation of the underlying data, not of the label logic, and is why "Active/Ongoing" is deliberately left Unknown rather than guessed either way.

15. Data Quality Limitations

1. No qualitative business-model or NLP free-text fields (company_description, problem_solved, solution_description, target_audience, technology_used, business_model, revenue_model, pricing_model, revenue_stream, employee_count) exist in the underlying Crunchbase snapshot exports - they are 100% Unknown for records sourced from the two Kaggle Crunchbase exports, disclosed per-column rather than filled in.
2. major_investors and last_funding_amount_usd are only populated for the 25 records from the 2024-2025 named-company source; the larger Crunchbase exports report lifetime funding totals and round counts, not per-round dates, amounts, or investor names.
3. acquisition_date / ipo_date / shutdown_date are Unknown for all Crunchbase-export records - the source provides a single status flag only, not the date that status was reached.
4. market_size_category / market_growth_category / demand_level are Unknown by design across the entire dataset - no external market-research or TAM source was used. The Market_Data sheet instead reports a clearly labeled derived statistic (count of same-industry, same-country startups within this dataset), which is not an external market-size figure.
5. The two Crunchbase exports are historical snapshots, not live data. A company shown as 'operating' may have changed status since the snapshot was taken - this is why an 'operating' flag is never treated as a success signal.
6. industry classification is a deterministic, rule-based keyword mapping over the source's category tags, not a human-verified ground-truth label; ambiguous or missing tags fall into 'Other / Unspecified' (13,665 of 67,021 records in the full pool).
7. Deduplication was conservative: only exact name + country + founded_year matches were merged (102 records). Same-name/same-country pairs with a different or missing founded_year were flagged, not removed, since generic names can legitimately belong to different real companies.
8. The real, cleaned pool (67,021 records) exceeds the ~5,000 requested. Rather than truncate arbitrarily, the main deliverable is a reproducible stratified sample (random_state=42); the full pool is shipped separately as Full_Cleaned_Pool_67021_records_BONUS.csv.

6. Sources

1. Big Startup Success/Fail Dataset (Crunchbase) - Kaggle, uploader yanmaks - kaggle.com/datasets/yanmaks/big-startup-secsees-fail-dataset-from-crunchbase
2. StartUp Investments (Crunchbase) - Kaggle, uploader arindam235 - kaggle.com/datasets/arindam235/startup-investments-crunchbase
3. Startups & Funding Dataset (2024-2025) - Kaggle, uploader sufya6 - kaggle.com/datasets/sufya6/startups-and-funding-dataset-20242025

Full field-by-field provenance is in the Data_Dictionary and Sources sheets of LaunchLens_Master_Dataset.xlsx.